Mark Li

mkli627554@gmail.com | mark-li.com | linkedin.com/in/mkli | github.com/marker6275 | +1 (832) 945-8288

EXPERIENCE

Shure Inc. *Chicago, IL (Remote)*
Applications Development Engineer Jul. 2025 - Aug. 2026
Applications Development Engineer Intern Jun. 2024 - Mar. 2025

- Eliminated manual reporting and saved 5+ hours weekly for 60+ engineers by building an internal metrics platform aggregating CI, testing, and operational data across 7+ services
- Improved reliability of real-time device synchronization across distributed conferencing hardware deployments by developing production frontend systems for a large-scale enterprise audio management platform
- Enabled low-latency synchronization and monitoring across network-connected audio devices by integrating GraphQL-based real-time device state pipelines into the frontend
- Reduced engineering review bottlenecks by 50% building an internal hackathon tool that dynamically assigned PR reviewers based on repository ownership and reviewer load balancing

Northwestern University - McCormick School of Engineering *Evanston, IL*
Teaching Assistant, Programming Languages (CS321) Sep. 2023 - Mar. 2024

- Debugged and evaluated student implementations of interpreters and language features in Racket (PLAI), guiding a cohort of 70+ students through semantic and implementation errors

Explorate (Startup) *Evanston, IL*
Full-Stack Developer Jun. 2023 - Sep. 2023

- Built a full-stack operations platform using Vue.js, Go, and PostgreSQL used by 80+ volunteer teams globally to coordinate volunteering opportunities
- Designed PostgreSQL schemas and optimized query patterns supporting scalable CRUD workflows and low-latency data access

PROJECTS

Verdra | Static analysis tool for cost-aware code optimization

- Built and deployed a static analysis platform using AST and control-flow analysis to identify cloud cost and performance inefficiencies, including redundant database calls, oversized imports, and over-provisioned serverless workloads
- Implemented automated rule evaluation pipelines capable of detecting 10+ optimization patterns across real-world codebases to reduce infrastructure waste and developer debugging overhead

RESEARCH EXPERIENCE

Northwestern University - McCormick School of Engineering
Embodied Systems Lab
Graduate Researcher Mar. 2024 - Jun. 2025

- Built fault-tolerant execution systems on MSP430 microcontrollers capable of recovering from intermittent power loss and constrained-memory failures in embedded environments
- Engineered low-overhead embedded recovery mechanisms under strict energy and memory constraints to maintain persistent execution state across device interruptions
- Developed and led a workshop on tradeoffs in electronics and computer systems, illustrating how hardware constraints shape computation, storage, and energy use

Programming Languages Lab
Research Assistant Jun. 2022 - Mar. 2024

- Evaluated probabilistic contract enforcement strategies across 13+ benchmarks to analyze performance-correctness tradeoffs in Racket

EDUCATION

Northwestern University Jun. 2025
Master of Science in Computer Science *Evanston, IL*
Bachelor of Science in Computer Science
Bachelor of Arts in Music

SKILLS

Programming Languages: Python, TypeScript, JavaScript, C/C++, Go, Java
Frontend: React, React Native, Angular
Backend: Node.js, GraphQL, FastAPI, REST APIs, PostgreSQL, MongoDB, AWS
Languages: English (Native), Mandarin (Professional), Spanish (Conversational)